DEVSECOPS ENGINEERING SQUAD

ENTERPRISE CLOUD

AWS Cloud Architecture • Automated CI/CD • KMS Encryption • Disaster Recovery

Client: ASO Systems
Engagement: 40 hrs/week Dedicated Squad
Cutover Target: October 2027
Blended Rate: $55.00 / hr

Production AWS DevSecOps & Enterprise Reliability Partnership

A senior, battle-tested 3-engineer cloud squad providing end-to-end infrastructure engineering, immutable IaC deployments, container orchestration (ECS / Fargate / Docker), cryptographic data protection (AWS KMS / Secrets Manager), and automated cross-region disaster recovery to guarantee operational readiness for ASO Systems' October 2027 production cutover.

AWS Core: **ECS • Fargate • EC2 • RDS** | IaC & CI/CD: **Terraform • GitLab CI • GitHub Actions**

Security: **AWS KMS • IAM RBAC • ClamAV • Secrets Manager** | Compliance: **GDPR & Korean PIPA Controls**

Availability: **40 Hrs/Wk • Contract-to-Hire**

1. EXECUTIVE SUMMARY & SQUAD DELIVERY MODEL

3 Specialized Engineers, 1 Unified Rate, Zero Skill Gaps

The Engineering Squad Advantage

COST & SPEED

Enterprise cloud operations require diverse competencies: cloud systems architecture, low-level database tuning, automated deployment pipelines, and compliance governance. Rather than relying on a single engineer who must divide attention, ASO Systems receives a **coordinated 3-person squad** billed at a single, blended hourly rate:

- **Continuous Execution:** Parallel workstreams across Terraform IaC, container pipelines, and database replication without context switching.
- **Enterprise Resilience:** Eliminates single-point-of-failure risk; knowledge is shared across the squad with strict runbook governance.
- **Optimized Cost:** Full access to a Lead Architect, Cloud Database Engineer, and dedicated Project Manager at **$55.00/hr** (within your $50–$70/hr range).

Direct Alignment with ASO Systems Mandate

MISSION SCOPE

Our squad directly maps to every technical objective outlined in your enterprise specification:

- **Zero-Trust AWS Foundation:** Complete isolation of Dev, Staging, and Production VPCs with strict IAM least-privilege boundaries and AWS Secrets Manager.
- **Automated Disaster Recovery:** Cross-region PostgreSQL/RDS replication, automated health checks, controlled writer promotion, and data reconciliation routines.
- **Cryptographic Data Lifecycle:** Application-layer encryption with AWS KMS envelope keys, automated annual rotation, and regulatory retention/secure deletion.
- **October 2027 Cutover Governance:** Rigorous load testing, simulated DR failovers, runbook authoring, and seamless transition to internal operations.

2. TECHNICAL CAPABILITY ALIGNMENT MATRIX

Mapping ASO Systems Requirements to Proven Implementation Tools

ASO SYSTEMS OPERATIONAL REQUIREMENT	SQUAD IMPLEMENTATION STRATEGY & TOOLING	VERIFIED REAL-WORLD PRECEDENT
Containerized Infrastructure (ECS / Fargate)	Terraform modular IaC managing Amazon ECS task definitions, Fargate serverless compute, Application Load Balancers (ALB), and auto-scaling triggers.	Multi-tenant e-commerce platform & Adrik microservices containerized with Docker and orchestrated in production.
Automated CI/CD & Deployment Controls	Multi-stage pipelines (GitLab CI / GitHub Actions) with SonarQube quality gates, container image vulnerability scanning, manual approvals, and instant rollbacks.	DIMO Lanka Gate Pass CI/CD with SonarQube & JaCoCo gates; automated deployment across Dev, UAT, and Production.
Database Operations & Cross-Region DR	Amazon RDS (PostgreSQL/MySQL) multi-AZ setup, cross-region read replicas, automated snapshot management, and controlled writer promotion scripts.	End-to-end production data migration and continuous RDS operations for SINGER Sri Lanka PLC financial workflows.
Security, KMS Keys & Secrets Management	AWS KMS envelope encryption with annual customer-managed key (CMK) rotation, AWS Secrets Manager dynamic credential injection, and IAM policy auditing.	Keycloak OAuth2/OIDC token verification middleware, AWS S3 file encryption with automated ClamAV antivirus scanning.
Privacy & Regulatory Controls (GDPR / PIPA)	Structured data retention lifecycles, cryptographic erasure protocols, immutable audit event logs via Kafka/CloudTrail, and PII anonymization pipelines.	Cross-system data governance on Castle Visualizer and Arimac eCom SaaS enterprise client platforms.

Immediate Availability & Long-Term Roadmap

Our squad is immediately available for 40 hours per week under an ongoing contract-to-hire arrangement. We are committed to supporting ASO Systems through all critical development sprints, DR recovery drills, the October 2027 production cutover, and subsequent operational acceptance.

MEET THE DEVSECOPS SQUAD

ROLES & EXPERTISE

Complementary Engineering Specializations Dedicated to ASO Systems

Engagement Model: Squad-as-a-Service
Lead Contact: Viranga Lakshan
Timezone: UTC+5:30 (Extensive Global Overlap)

3. SQUAD ENGINEERING PROFILES

Lead Cloud Architect + Cloud Infrastructure/DBA + Dedicated Project Manager

Viranga Lakshan

LEAD CLOUD ARCHITECT & DEVSECOPS LEAD

Associate Software Engineer, Arimac Lanka • 4+ Years Enterprise Experience

Leads AWS cloud architecture, containerized infrastructure (ECS/Fargate/Docker), Terraform IaC, and security engineering. Expert in zero-trust networking, Keycloak/OIDC identity management, and automated CI/CD pipeline governance with automated security scanning.

- Architected microservices with Docker, Kubernetes, and GitLab CI/CD for international media & e-com platforms.
- Engineered AWS S3 secure file ingestion with ClamAV antivirus scanning and Spring Kafka audit events.
- Authored multi-environment Terraform configs and SonarQube/JaCoCo automated pipeline gates.

AWS ECS/Fargate | Docker | Terraform | GitLab CI / Jenkins | Keycloak / OIDC | AWS S3 / ClamAV

Janith Kavishka

CLOUD INFRASTRUCTURE & DATABASE ENGINEER

Programmer, DMS Electronics • Enterprise Core Systems & Database Operations

Specializes in production database operations, Amazon RDS management, message-driven microservices (RabbitMQ, Kafka), and serverless execution. Hands-on experience executing live enterprise database migrations and high-availability operational support.

- Designed and executed full data migration to Amazon RDS (MySQL) for SINGER Sri Lanka PLC financial platform.
- Built enterprise stock-flow and tracking platform integrated with AWS Lambda and Amazon RDS.
- Implemented automated DB backup routines, point-in-time recovery, and multi-tier reconciliation.

Amazon RDS | AWS Lambda | PostgreSQL / MySQL | RabbitMQ / Kafka | AWS EC2 | DB Recovery & DR

Lithumitha Ravikumar

PROJECT MANAGER

Associate Project Manager, Arimac Lanka • Enterprise SaaS & Data Platforms

Leads Agile project management, sprint ceremonies, delivery coordination, release governance, and quality assurance. Coordinates cross-functional workflows across engineering, QA, and business stakeholders, ensuring milestone adherence, risk mitigation, and delivery excellence.

- Managed 4 engineering squads delivering enterprise multi-tenant digital commerce SaaS platforms.
- Led delivery of Castle Visualizer enterprise data analytics platform for Australian corporate client.
- Coordinates sprint ceremonies, risk management, QA acceptance validation, and delivery documentation.

Project Management | Sprint Coordination | Release Governance | Risk Management | QA Acceptance | Stakeholder Reporting

4. CORE AWS DEVSECOPS & SECURITY PILLARS

Enterprise Standards Applied Across All ASO Systems Environments

Infrastructure as Code & Automated Pipelines

TERRAFORM & CI/CD

Immutable Infrastructure: 100% of AWS resources (VPCs, ECS task definitions, security groups, RDS parameter groups) declared in modular Terraform with remote state locking via S3 and DynamoDB.

Multi-Stage CI/CD Security: GitLab CI / GitHub Actions workflows enforcing static code analysis (SonarQube), container CVE scanning (Trivy), automated integration tests, manual staging-to-production approval gates, and zero-downtime blue/green rollbacks.

Disaster Recovery & Database High Availability

RTO < 30M • RPO < 5M

Cross-Region DR: Multi-AZ deployment in primary AWS region with cross-region asynchronous read replicas. Automated CloudWatch alarms trigger controlled database writer promotion runbooks.

Data Reconciliation: Post-failover reconciliation scripts verifying transaction sequence numbers, resolving split-brain scenarios, and ensuring data integrity before opening client traffic.

Cryptographic Security & KMS Key Lifecycle

AWS KMS & SECRETS

Envelope Encryption: Sensitive data encrypted at application level using AWS KMS Customer Master Keys (CMK) and AES-256 GCM data keys. Secrets dynamically injected via AWS Secrets Manager with zero hardcoded credentials.

Automated Rotation: Seamless annual key rotation and re-encryption workflows with cryptographic audit logs streamed directly to AWS CloudTrail.

Regulatory Data Compliance & SRE Monitoring

GDPR & PIPA CONTROLS

Compliant Data Lifecycle: Automated S3 lifecycle rules and RDS snapshot pruning enforcing GDPR / Korean PIPA data retention mandates, cryptographic deletion, and immutable audit logs.

Centralized Observability: CloudWatch metric dashboards, VPC Flow Logs, CloudTrail security auditing, anomaly alerting via PagerDuty/Slack, and detailed operational runbooks.

VERIFIED PRODUCTION TRACK RECORD

Enterprise AWS, Containerization, Database Operations, and Release Governance

REAL PROJECTS

Clients Served: Enterprise, Multinational, Gov
Architecture: Microservices & Event-Driven
Compliance: High-Security & Enterprise Audit

5. ENTERPRISE CASE STUDIES & TECHNICAL PRECEDENTS

Production Systems Built and Operated by Squad Members

DIMO Gate Pass Management System

Client: DIMO Lanka (Enterprise Automobile & Logistics) • Lead: Viranga Lakshan

SECURITY & CI/CD

Architecture & Infrastructure: Mission-critical enterprise vehicle dispatch and gate security platform built as high-throughput Spring Boot microservices with Spring Kafka event streaming, Keycloak OAuth2/JWT security, and MySQL persistence with Flyway migrations.

- **Secure File Handling with ClamAV & AWS S3:** Architected secure file-upload ingestion pipeline to AWS S3 with automated asynchronous ClamAV virus scanning, quarantine buckets, and IAM role-delegated presigned URLs.
- **Containerization & CI/CD Quality Gates:** Containerized microservices using Docker; implemented robust Jenkins and GitLab CI/CD pipelines integrating automated SonarQube static code quality analysis and JaCoCo test coverage verification before artifact promotion.
- **Multi-Role Approval Workflows:** Built role-based access control (RBAC) and audit event publishing across SAP, Azure AD / Microsoft Graph, Google Maps API, and internal security personnel.

Tech Stack: Docker, AWS S3, Jenkins, GitLab CI/CD, Keycloak, Spring Kafka, MySQL Flyway, ClamAV, SonarQube, JaCoCo, Java 21 / Spring Boot.

Enterprise Financial Operations & Amazon RDS Migration

Client: SINGER Sri Lanka PLC (DMS Electronics) • Lead: Janith Kavishka

AMAZON RDS & LAMBDA

Database Operations & Cloud Migration: Engineered production database migration to **Amazon RDS (MySQL)** for an island-wide financial field collection, batch-wise transaction tracking, and mobile receipt generation platform.

- **Zero-Downtime Migration & Data Integrity:** Managed schema re-architecting, continuous data replication, and high-volume reconciliation during cutover, ensuring zero data loss across millions of financial records.
- **AWS Serverless & Event Streaming:** Built multi-tier distribution tracking platform utilizing **AWS Lambda** integrated with Amazon RDS and Spring Boot backend microservices using RabbitMQ for reliable, message-driven transaction processing.
- **Production DBA Support:** Ongoing query optimization, automated snapshot lifecycle scheduling, index tuning, and uninterrupted operational support for enterprise clients.

Tech Stack: Amazon RDS, AWS Lambda, Docker, Java / Spring Boot, RabbitMQ, React, Next.js, MySQL, Enterprise Backup & Replication.

Adrik News & E-Com SaaS

Client: Adrik (Kuwait) & Arimac

KUBERNETES & OIDC

Bilingual Media & Multi-Tenant SaaS: Backend microservices connecting web and mobile applications to core APIs. Containerized with Docker and deployed to **Kubernetes** via GitLab CI/CD across Dev, UAT, and Production.

- Keycloak OAuth2/OIDC authentication with JWT validation middleware and token refresh routines.
- Multi-tenant database schema isolation and dynamic tenant provisioning.

Castle Visualizer & Arimac eCom

Client: Castle Group (Australia) • PM: Lithumitha

RELEASE GOVERNANCE

Enterprise Data Platform & SaaS Delivery: Multi-dashboard analytics platform consolidating data across legacy ERP and operational business systems into real-time reporting plane.

- Sprint planning and release governance across 4 engineering squads with strict change management.
- Enforced rigorous QA acceptance criteria, audit compliance tracking, and incident post-mortems.

EXECUTION BLUEPRINT & CUTOVER ROADMAP

Structured Phasing to Ensure Production Hardening, Disaster Recovery, and Operational Readiness

OCTOBER 2027

Commitment: 40 hrs/wk Dedicated
Target Cutover: October 2027
Rate: $55.00/hr Squad Total

6. PHASED TECHNICAL EXECUTION BLUEPRINT

From Baseline IaC Hardening to October 2027 Production Cutover & Handover

Phase 1: Environment Isolation & IaC Foundation

MONTHS 1–3

- Audit current infrastructure and codify all resources into clean, modular **Terraform** scripts with state locking in Amazon S3/DynamoDB.
- Establish strictly isolated VPCs for Development, Staging, and Production with public/private subnet topology and NAT gateways.
- Implement AWS IAM role-based least privilege, enforce MFA for console access, and configure AWS Secrets Manager for all environment credentials.
- Deploy containerized base environments using Docker on Amazon ECS / AWS Fargate behind Application Load Balancers.

Phase 2: Secure CI/CD & KMS Key Lifecycles

MONTHS 4–7

- Build automated CI/CD pipelines (GitLab CI / GitHub Actions) integrating SonarQube SAST, Trivy container security scans, and automated unit test gates.
- Implement AWS KMS envelope encryption for sensitive database columns and S3 artifacts with automated annual CMK rotation.
- Configure centralized observability: AWS CloudWatch container insights, CloudTrail audit logs, and proactive PagerDuty alerting.
- Establish GDPR and Korean PIPA data retention lifecycles, automated backup archiving, and secure cryptographic deletion scripts.

Phase 3: Database High Availability & DR Drills

MONTHS 8–14

- Deploy Amazon RDS (PostgreSQL/MySQL) multi-AZ primary cluster with automated asynchronous cross-region read replica for disaster recovery.
- Script automated DR failover workflows: controlled writer promotion, DNS routing updates via Route 53, and application reconnection pooling.
- Conduct quarterly simulated DR exercises to test cross-region recovery, measure RTO/RPO, and validate post-failover data reconciliation routines.
- Optimize database performance: connection pooling, query indexing, slow-query logging, and storage auto-scaling.

Phase 4: Load Testing, Cutover & Handover

MONTHS 15–18+ (OCT 2027)

- Lead comprehensive pre-cutover performance and load testing simulating peak enterprise transaction volumes on ECS and RDS.
- Coordinate third-party penetration testing, vulnerability remediation, and security posture sign-off.
- **Lead October 2027 Production Cutover:** Execute synchronized cutover runbook with zero downtime, live data verification, and hypercare standby.
- Deliver comprehensive operational runbooks, disaster recovery procedures, and conduct structured handover sessions for internal teams.

7. ENGAGEMENT TERMS & OPERATIONAL SLA

Predictable, Transparent Enterprise Partnership

Dedicated Bandwidth

COMMITMENT

40 Hours / Week: Continuous coverage across development, testing, recovery drills, and cutover. Direct participation in daily standups and sprint planning.

Single Blended Rate

COST CERTAINTY

$55.00 / Hour: Transparent hourly billing covering the entire 3-person squad under a single Upwork contract-to-hire agreement, with zero hidden fees.

Cross-Functional Synergy

COLLABORATION

Active, collaborative integration with ASO Systems' software architects, .NET developers, QA engineers, and external security compliance auditors.

Next Steps & Technical Discovery Call

We welcome a technical discussion with ASO Systems engineering leadership to review your current AWS architecture, evaluate your October 2027 cutover milestones, and present our Terraform and DR runbook templates. We are ready to begin immediately upon award.